Panskura Banamali College (Autonomous)

B.Sc.(H) 1st Semester Examination 2022 (Online mode)

Subject : BCA **Paper: CC2T**

Paper Name : Computer Fundamental with Digital Electronics

F.M. : 40 **Time : 2 Hours**

Answer any four questions: **4*10=40**

1. a) Find 8-bit binary equivalent of $(120)_{10}$ 2
b) Find the gray code equivalent to binary number 110101. 2
c) How does a super computer differ from a personal computer? 2
d) Write an algorithm that detects if a number read from keyboard is zero or not. 4

2. a) Show the excess-64 code of binary number 11011 in an 8-bit register. 2
b) Consider the Boolean function f(a,b,c,d)= Σ(0,2,4,5,7,8,14,15). Minimize the Boolean function using Karnaugh Map. 4
c) Design a adder-cum-subtractor circuit that adds/subtract two bits depending on choice of control signal. 4

3. a) What are the difference between decoder and demultiplexer ? what is universal shift register? 2+3
b) What do you mean by priority encoder ? What are the difference between synchronous and asynchronous sequential circuits ? 2+3

4. a) How does sequential circuit differ from combinational circuit? 2
b) What is the disadvantage of J-K flip flop? How can we overcome this disadvantage? Give circuit diagram of master slave J-K flip flop. 2+2+4

5. a) Why we used k-map for Boolean expression. 2
b) Briefly describe about half adder and full adder . 5
c) solved that : a: $(57)_8 + (54)_8$ b: $(BCA)_{16} - (A2)_{16}$ 3

6. a) prove that X-OR gate is a buffer. 2
b) Briefly describe about full substractor . 5
c) why we use 2's complement for binary subtraction. 2
d) solved that $(-15)_{10}-(12)_{10}$ using 2's complement. 1

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